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Mechanism of wettability enhancement of the AgCuTi-SiO₂f/SiO₂ system with a laser-manufactured bioinspired sinusoidal interface

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Metrics

Outline

Highlights

Abstract

Graphical abstract

Keywords

1. Introduction

2. Experimental

3. Results

4. Discussion

Highlights

- Bioinspired sinusoidal structures were established in the SiO₂f/SiO₂ interface.
- Reactive wetting was promoted on the sinusoidal surface due to the active defects.
- The double sinusoidal structure achieves the best wettability and joint strength.
- The contact angle reduced from 110° to 70°, joint strength rose from 5MPa to 23MPa.

Abstract

The poor wettability of filler metals on the SiO₂_f/SiO₂ composite surface adversely affected the brazing joint strength, constraining the application of SiO₂_f/SiO₂ composite. In this paper, the bioinspired sinusoidal structure interface was designed and manufactured by the picosecond laser, aiming to enhance wettability. The wetting angles of the Ag-Cu-Ti filler metal on the SiO₂_f/SiO₂ composite surface with different bioinspired sinusoidal structures were compared. Effects of the structural morphology and surface reaction state of bioinspired sinusoidal structures on the wetting mechanism were investigated. The results indicated that, as the amplitude of the sinusoidal structure decreased, the surface energy barrier that hindered wetting and the capillary force that promoted droplet spreading decreased simultaneously. Furthermore, active defects, such as ODC(II) and NBOHC, were generated on the structured surface. The surface rich in defects easily reacted with the Ag-Cu-Ti filler metal, which was beneficial for promoting the reactive wetting. An optimized double sinusoidal structure with 25μm amplitude and 100μm wavelength reduced the wetting angle from ~110° to ~70°. Dense and continuous Ti₅Si₃+TiO₂/Cu₃Ti₃O reaction layers with a thickness of over 2μm were formed, which indicated a reliable metallurgical bonding. Attributed to the promoted wettability and enhanced interfacial bonding, the SiO₂_f/SiO₂ composites brazed joint was strengthened from 5MPa to 23MPa. This bioinspired sinusoidal structure could also be introduced in further ceramics-metal dissimilar interfaces to enhance wettability and interfacial bonding.

Graphical abstract

5. Conclusions

CRediT authorship contribution statement

Declaration of Generative AI and AI-assis...

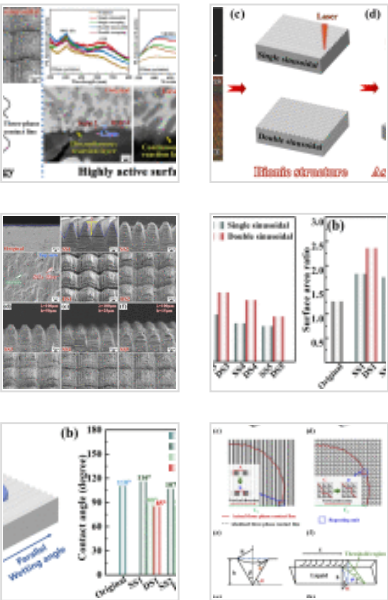
Declaration of competing interest

Acknowledgments

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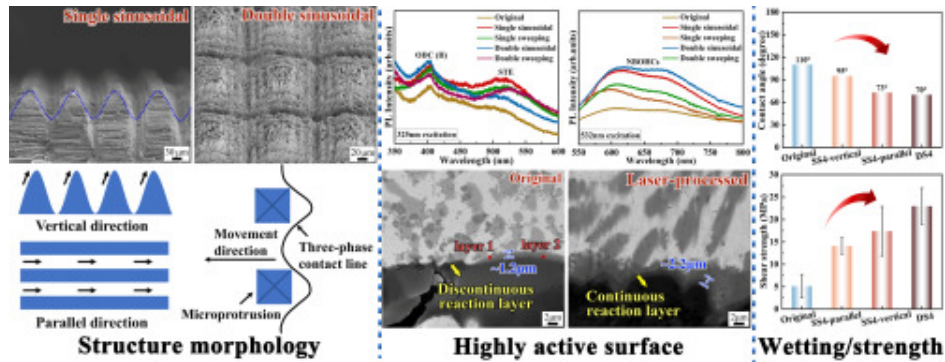
Figures (16)



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Tables (1)

Table 1. Average chemical compositions ...



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Keywords

Bioinspired sinusoidal interface; Laser-induced active defect; Surface energy barrier; Capillary driving force; AgCuTi-SiO₂f/SiO₂ system

[Previous article in this issue](#)

[Next article in this issue](#)

1. Introduction

SiO₂f/SiO₂ composites have emerged as promising candidate materials for critical components in new-generation ceramic systems due to their low density, high wave transmittance, excellent dielectric properties, and remarkable thermal shock resistance. These superior attributes have facilitated widespread adoption in aerospace, nuclear energy, and other high-performance engineering sectors, thereby driving technological advancement in these fields [1], [2], [3]. To further expand the application scope of the composite ceramics, it is essential to achieve reliable joints with other materials especially metallic parts. Among various joining methods, active brazing is considered the

effective method due to its operational simplicity and excellent wettability to inert ceramic surfaces [[4], [5], [6]]. However, the substantial mismatch in the coefficient of thermal expansion (CTE) between the SiO₂f/SiO₂ composite and the metals generates substantial residual stresses within the joint, which can lead to crack propagation and joint failure [[7], [8], [9], [10]].

Various surface modification techniques have been proposed to release residual stress, such as the introduction of whiskers in the brazing joint [11,12]. However, they tend to react with the filler metal to form brittle phases. Alternatively, the wettability could be improved by selectively etching the less wettable SiO₂ matrix using HF acid [13], but this will damage the performance of the base material. In addition, the laser drilling and grooving methods [14,15] have been employed to fabricate interface microstructure for stress dispersion, but the influence of microstructure morphology on residual stress has not been thoroughly studied. Given these limitations in existing methods, there is an urgent need to propose an innovative and effective method to alleviate the residual stress at the joints between ceramic composites and metals.

The survival competition in nature has driven the evolution of diverse structures and interface architectures of biological materials, giving rise to various strengthening and toughening mechanisms [16,17]. In particular, mantis shrimp has become a focal point of research in biology and materials science due to the exceptional fracture toughness and impact resistance of its dactyl club [18,19]. The striped region of the dactyl club features a wavy interface between hard phase mineralized fibers and the soft phase organic matrix, which facilitates stress transfer along the interface and mitigates stress concentration [20,21]. Inspired by the structural characteristics of the striped zone, bioinspired sinusoidal structures are fabricated on the SiO₂f/SiO₂ composite by ultrafast pulse laser to alleviate the residual stress and enhance the joint strength. However, there are still two challenges to introducing micro-nanostructure at the bonding interface. Firstly, Li et al. found that the filler metal could not completely fill into the microstructure with greater groove depths, which indicates the need to investigate the wetting and filling behavior on structured surfaces as a prerequisite for reliable joints [22]. Secondly, Lößlein et al.

discovered that laser patterning can induce both morphological changes and chemically modified active surfaces [23], the influence of such active surfaces on interfacial reaction and high-temperature wetting behavior remains unexplored. Therefore, it is crucial to independently study the influence of sinusoidal structural morphology and active surface on wettability.

This study systematically investigates the influence of bioinspired sinusoidal structures on the wettability of filler metals from two perspectives: structural morphology and active surface. Firstly, a thermodynamic model was established to explore the physical mechanisms governing wetting and spreading behavior of the filler metal perpendicular to the single and double sinusoidal structures, based on the relationship between variations in surface energy barrier and the instantaneous wetting angle. Simultaneously, a wetting kinetic model of filler metal within a single sinusoidal structure groove was developed to investigate the driving role of capillary forces on wetting behavior along the groove direction. Secondly, to eliminate the impact of structural morphology on wettability, sweeping samples are prepared to compare with the original surface in order to investigate the effect of laser-induced active surfaces on wettability.

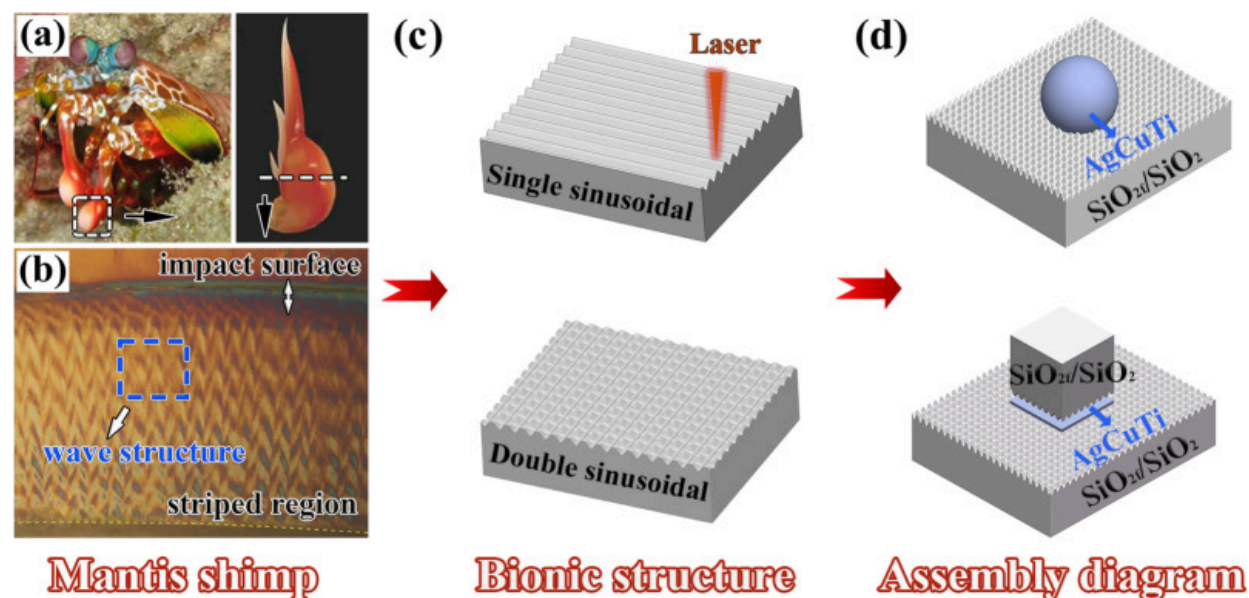
2. Experimental

2.1. Materials

In this study, SiO₂f/SiO₂ composite supplied by the Institute of Ceramics featured a layered woven structure composed of interlaced fibers (warp and weft) within a matrix. Then SiO₂f/SiO₂ composites were sectioned into 10mm×10mm×3mm blocks for wetting experiments, while two distinct sizes (4mm×4mm×3mm and 10mm×10mm×3mm) were prepared to serve as the upper and lower joining partners for brazing experiments. Prior to processing, all specimens were ultrasonically degreased with acetone.

2.2. Bioinspired sinusoidal interface preparation

As shown in Fig. 1a-b, these are the structure characteristics of the mantis shrimp and its striped region, where the interface between the mineralized fibers and the organic matter forms a continuous wavy connection (simplified as a sinusoidal curve). Given the hard and brittle characteristics of SiO₂f/SiO₂ composites, a picosecond laser (PX200-3-GF, Edge Wave, Germany) with a maximum output power of 30W and a spot size of 10μm was selected to fabricate bioinspired sinusoidal structures. Furthermore, the Gaussian distribution characteristic of the laser beam required the use of a multi-track scanning strategy to achieve uniform structures. Fig. 1c presents the schematics of the processed single and double sinusoidal structures. These structures were fabricated employing symmetric scanning speed groups: 300–400mm/s, 500–600mm/s, 800–900mm/s, 900–1000mm/s and 1400–1500mm/s to achieve five different amplitudes, with the resultant structures designated SS1-SS5 (single sinusoidal) and DS1-DS5 (double sinusoidal) in order of increasing scanning speed. Additionally, to accurately evaluate the independent effect of laser-scanned surfaces on wettability, the single and double sweeping structures were performed at 900-1000mm/s, named SS6 and DS6, respectively. Following laser treatment, the samples were ultrasonic washed and dried again.



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Fig. 1. (a) Mantis shrimp and dactyl club, (b) striped region in dactyl club [21], (c) bioinspired single and double sinusoidal structure schematics, (d) assembly diagram showing wetting of AgCu-4.5wt.%Ti spherical paste deposited on structured SiO₂f/SiO₂ surface and the structured SiO₂f/SiO₂-SiO₂f/SiO₂ joints brazed with AgCu-4.5wt.%Ti paste.

2.3. Wetting and brazing experiments

The Ag-21Cu-4.5Ti (wt.%) braze filler was mixed with cellulose solvent to prepare a composite braze paste. This prepared paste was then vertically deposited onto the sinusoidal structured SiO₂f/SiO₂ surface using tweezers for the wetting experiment (Fig. 1d). Subsequently, a controlled 200μm thick layer of the braze paste was uniformly distributed between the two sinusoidal-structured SiO₂f/SiO₂ composite blocks for brazing experiments. To optimize interfacial bonding, a 0.6Pa compressive load was subsequently applied to ensure intimate contact between the joining surfaces (Fig. 1d). Untreated original SiO₂f/SiO₂ composites served as controls for both experiments. The brazing process was conducted under a vacuum pressure of 1.0×10^{-3} Pa with isothermal holding at 880°C for 10min to optimize the flow characteristics of the brazing filler. Meanwhile, the dynamic wetting angle change was measured and recorded at 880°C using a vacuum dynamic wetting angle system (VAF-30) during the 60min holding period.

2.4. Characterization

A systematic characterization approach was implemented to evaluate the SiO₂f/SiO₂ composites before joining. Phase transformations following laser processing were first identified through X-ray diffraction analyzer (XRD). Optical properties were then quantified via ultraviolet/visible/near infrared absorption spectrometer (Lambda 950) to determine absorption coefficients across both original and laser-modified surfaces. Concurrently, structural reorganization of SiO₂f/SiO₂ composite before and after laser scanning were studied by using 785nm excitation beam induced micro-Raman

spectroscopy (inVia-Reflex). Subsequently, photoluminescence (PL) spectra analysis of the original surface and the laser processing regions was characterized using a confocal Raman spectrometer (inVia-Reflex and Horiba LabRAM Odyssey) with excitation wavelengths of 325 nm and 532 nm, respectively. While the macroscopic morphology of sinusoidal structures was simultaneously analyzed using the three-dimensional laser microscope (OLS5000 series, Olympus Corporation, Japan) with its 0.5 nm vertical resolution enabling precise measurement of surface area ratios and roughness parameters. After the brazing experiment, static wetting angles were characterized using a wetting angle measurement instrument (Dataphysics OCA20LHT). Microstructural evolution across both the wetting interface and the brazed joint was characterized using a Zeiss 560 field emission scanning electron microscope (SEM) equipped with energy-dispersive X-ray spectroscopy (EDS). To further characterize the in-situ reaction products, TEM analysis was performed using a transmission electron microscopy (TEM, TALOS F200X). Finally, joint shear strength was measured using a universal testing machine at a constant crosshead speed of 0.1 mm/min, with the average value determined from at least five brazed specimens tested under identical conditions.

3. Results

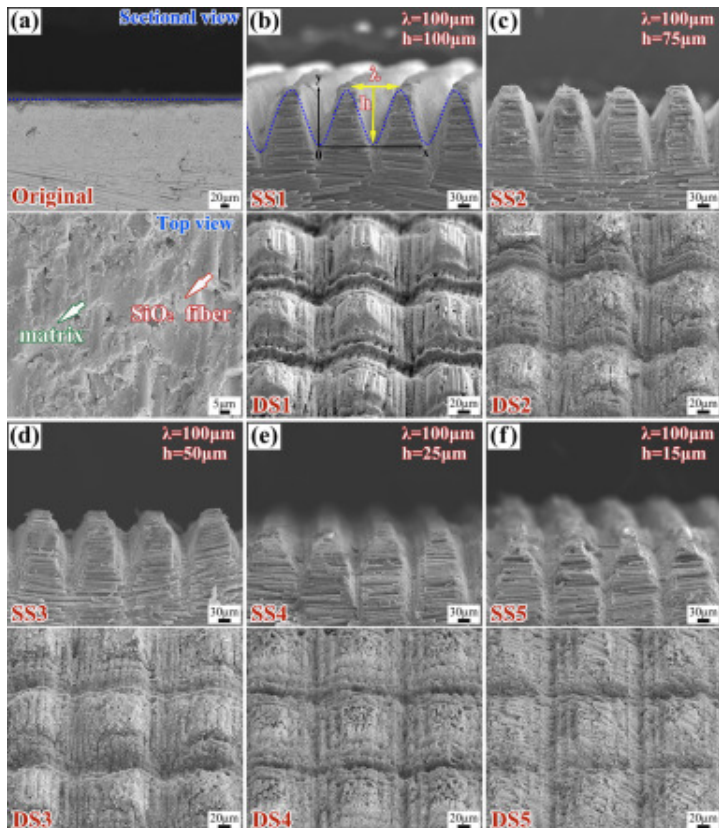
3.1. Sinusoidal structural morphology

Prior to laser processing, the SiO₂_f/SiO₂ composites are layered woven structures composed of interlaced fibers (warp and weft) and matrix. As shown in Fig. 2a, the cross-section of the SiO₂_f/SiO₂ composite was relatively flat, while the top view revealed the SiO₂ matrix filling the inter-fiber spaces and partially covering individual fibers. Bioinspired single sinusoidal structures, resembling the wave morphology of the mantis shrimp striped region, were fabricated on SiO₂_f/SiO₂ composite at different scanning speed groups (upper views of Fig. 2b-f). Moreover, the amplitudes of the sinusoidal structures gradually decreased from 100 μm to 15 μm with increasing laser scanning speed group. Then the relationship between scanning speed groups and the structure

dimensions was established by analyzing the effective accumulated fluence per area of laser irradiation on the composite surface. The equation for the effective accumulated fluence per area F can be deduced as follows [24]:

$$F = \frac{P}{vl} \quad (1)$$

while P is the average power, l is defined as the distance between overlapping scan line and the v is the scanning speed. Under constant power and inter-track distance, the effective accumulated fluence is inversely related to scanning speed. High fluence corresponding to the lower speed group primarily affects the amplitude direction, leading to significant material removal. As the scanning speed increases, the corresponding accumulated fluence decreases, leading to the reduction in amplitude [25].

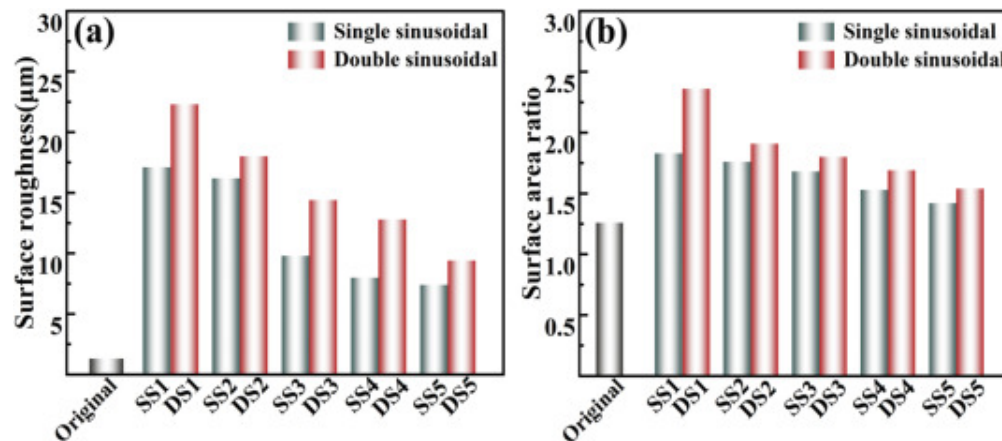


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Fig. 2. (a) Top and sectional views of the original SiO₂f/SiO₂ composite, and macroscopic morphologies of single and double sinusoidal structure at different scanning speed groups: (b) SS1 and DS1, (c) SS2 and DS2, (d) SS3 and DS3, (e) SS4 and DS4, (f) SS5 and DS5.

To achieve greater isotropy and uniformity, the double sinusoidal structures were fabricated by orthogonal processing using the same laser parameters (lower views of [Fig. 2b-f](#)). The ablation process induced the formation of micro-protrusions along the scanning path edges and micro-pits at the path intersections. With increasing laser scanning speed group, the amplitude of both the micro-protrusions and micro-pits decreased, following a trend consistent with that observed in the single sinusoidal structures. The macroscopic morphological differences between single and double sinusoidal structures were quantitatively characterized using surface roughness and surface area ratio. As shown in [Fig. 3](#), the sinusoidal structured surfaces exhibited higher values for both parameters than the original surface. This significant increase could be primarily attributed to the unique periodic geometry of the sinusoidal structures. Additionally, a comparison of the characteristic values between single and double sinusoidal structures further showed that the former exhibited lower values, with both parameters decreasing as the scanning speed increased. This phenomenon aligned with the previously mentioned trend of structural depth variation.



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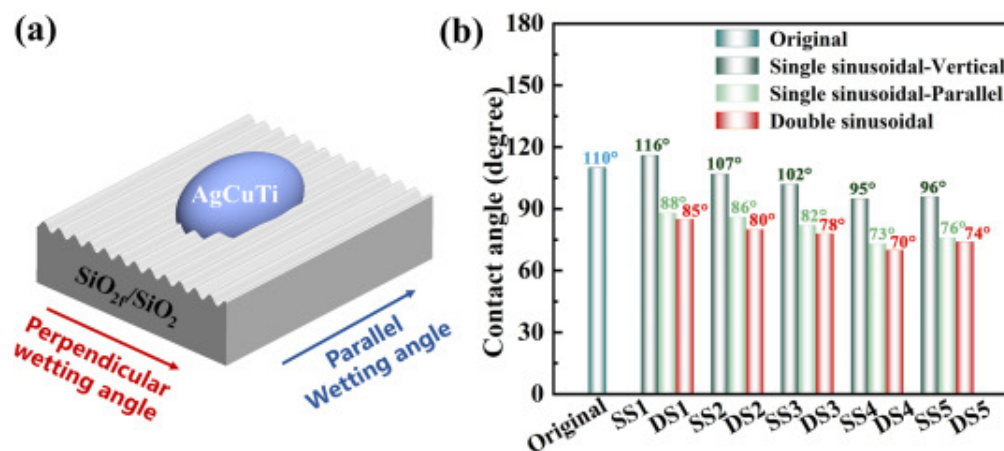
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Fig. 3. Surface topography parameters: (a) surface roughness values, (b) surface area ratio values.

3.2. Surface wettability

Fig. 4 shows the static wetting angles of the filler metal on the original and the sinusoidal structured surfaces fabricated with different scanning speed groups. The single sinusoidal structure exhibited pronounced anisotropic wetting behavior [26,27], with a distinct difference between the wetting angles measured in the perpendicular and parallel directions, as shown in Fig. 4a. The single sinusoidal structure SS1 exhibited a larger perpendicular wetting angle (116°) than the original surface (110°). This phenomenon indicated that deep micro-ridges introduced significant physical barrier effects. As the scanning speed increased from SS2 to SS5, the wetting angles generally decreased, reaching a minimum (95°) at SS4, before a slight increase to 96° at SS5. In comparison, the parallel wetting angles of the single sinusoidal structure were consistently lower than both those measured in the perpendicular direction and the original value. This suggests that the micro-groove provided an effective physical pathway for filler spreading, thereby enhancing wettability. The wetting angle in this direction also showed a general decrease

with increasing scanning speed group, reaching its minimum (73°) at SS4, followed by a slight increase at SS5. Notably, the double sinusoidal structure exhibited superior wetting performance compared to both directions of the single sinusoidal structure, as shown in Fig. 4b. This enhanced performance was accompanied by a similar wetting angle trend, reaching the optimum (70°) at DS4. This indicates that the double sinusoidal structure possesses a unique structural design advantage in synergizing and enhancing multiple wetting driving mechanisms. In the following section, the anisotropic wetting mechanism of the single sinusoidal structure and the reasons for the significant wetting angle reduction in the double sinusoidal structure will be thoroughly investigated, with a focus on structural morphology and active surface factors.



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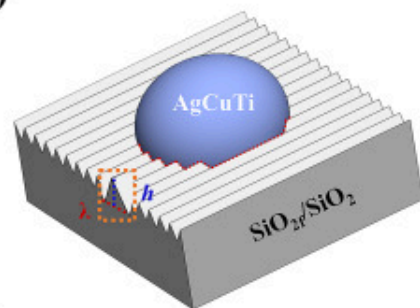
Fig. 4. (a) Schematic illustration of wetting angle observation directions for the single sinusoidal structure, (b) comparison of static wetting angles among the original surface, the single and double sinusoidal structure at different scanning speed groups.

3.3. Influence of structure morphology on wettability

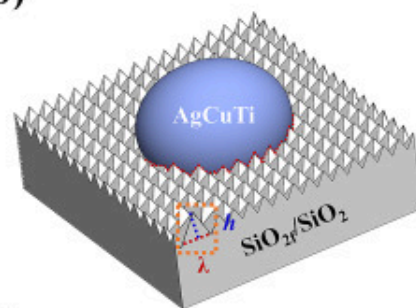
3.3.1. Surface energy barrier modeling

To investigate the physical barrier mechanism, a thermodynamic model was developed to analyze the brazing droplet spreading perpendicular to single and double sinusoidal structures. The model described the surface free energy barrier as a function of the instantaneous wetting angle, thereby revealing the influence of the micro-ridge, micro-protrusions, and their corresponding amplitude on the wettability. For geometric simplification, the single and double sinusoidal structures were modeled as a series of triangular prisms with isosceles triangles as their side faces (Fig. 5a) and a series of square pyramids (Fig. 5b), respectively, with the base length (λ) and height (h) representing the wavelength and amplitude of the structures. To enable comparison of the energy barrier between single and double sinusoidal structures, two approximations were made: firstly, the triple-phase contact lines are idealized as circular shapes, emphasizing their radii $L_{\perp}$ and $L'_{\perp}$ in the direction perpendicular to structure; secondly, repeating units of size $\lambda \times \lambda$ are selected for local analysis, as illustrated in Fig. 5c-d. Additionally, the following assumptions were introduced in the thermodynamic analysis to eliminate the influence of the droplet's intrinsic physical properties: (1) the tension and gravity of the triple-phase contact line are negligible. (2) the droplet surface is idealized as a spherical shape with conserved volume during spreading. (3) the Young's equation is locally valid, meaning the local curvature of the liquid-gas interface near the microstructure has a negligible impact on the system's energy. (4) the chemical reactions between the filler metal and SiO₂f/SiO₂ composite, and the laser-induced surface modification are also considered negligible.

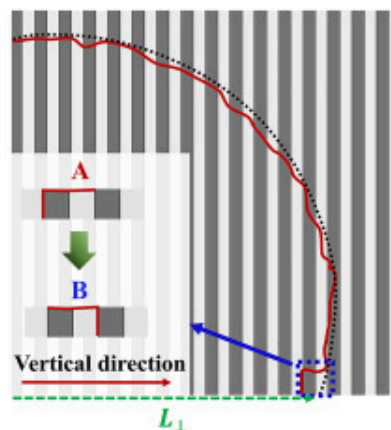
(a)



(b)



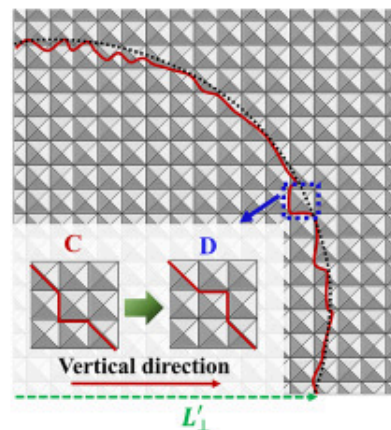
(c)



— Actual three phase contact line

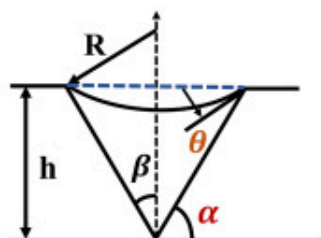
..... idealized three phase contact line

(d)

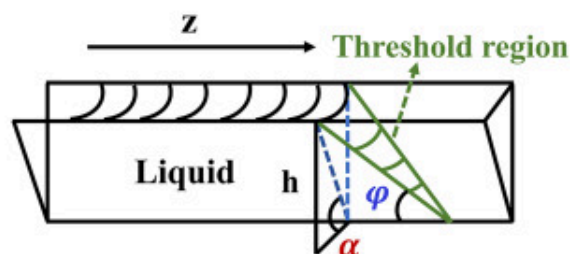


Repeating unit

(e)



(f)



(g)



(h)





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Fig. 5. (a) 3D model of a brazing droplet resting on the sinusoidal structures surface: (a) single sinusoidal structure (simplified as triangular prisms), (b) double sinusoidal structure (simplified as square pyramids), (c) variation of the three-phase contact line as the droplet moves through a single repeating unit from initial state A to state B on the single sinusoidal structure surface, (d) the variation of the three-phase contact line as the droplet advances from initial state C to state D across a single repeating unit on the double sinusoidal structure surface, (e)-(f) schematic diagram of brazing droplet spreading along the direction of single sinusoidal groove, (g) non-composite and composite wetting states, (h) cross-sectional schematic of a brazing droplet on the single sinusoidal structures surface.

Neglecting the effect of droplet gravity, the penetration of liquid into the grooves is primarily determined by capillary force. As shown in Fig. 5h, if the capillary force (F_c) concave menisci are formed, F_c drives the liquid to infiltrate the completely fill the grooves. Conversely, if the convex menisci are formed, F_c promotes the liquid to move outward from the grooves [28]. In the composite state, numerical analysis shows that the droplet basically exhibits isotropic wetting behavior on the rough surface [29]. Therefore, the wetting behavior on the sinusoidal structured surfaces in this study is described by the Wenzel model (Fig. 5g) [30].

The sinusoidal structure caused pinning of droplet's triple-phase contact line at the structural edges. This occurred when the system energy was insufficient to overcome the energy barrier imposed by the micro-ridges/protrusions, resulting in a metastable pinned state. Therefore, based on the dynamic characteristics and geometric of the actual contact

line, the variation of the surface energy barrier with the instantaneous wetting angle was described for the transition of the three-phase contact line from edge state A to adjacent edge state B in the single sinusoidal structure (Fig. 5c). Due to the local validity of Young's equation, the surface energy transition from state A to state B was expressed as:

$$\Delta F_{A-B}^N = 2\pi \frac{L_B^2}{1 + \cos\theta_B} - \frac{L_A^2}{1 + \cos\theta_A} - 2\lambda \sqrt{h^2 + \frac{\lambda^2}{2}} \cos\theta_Y \quad (2)$$

where the L_A and L_B represented the triple-phase contact line radius of the droplet in states A and B, respectively. The θ_A and θ_B represented the wetting angles of the droplet in states A and B, respectively, and the θ_Y was the intrinsic wetting angle. Based on the previous assumptions, the droplet volume remained constant during its movement, and the geometric constraint equation was shown as:

$$\frac{\pi L_A^2}{\sin\theta_A} \left(1 - \cos\theta_A - \frac{1}{3}1 - \cos\theta_A\right)^3 = \frac{\pi L_B^2}{\sin\theta_B} \left(1 - \cos\theta_B - \frac{1}{3}1 - \cos\theta_B\right)^3 + \frac{1}{2}\lambda^2 h \quad (3)$$

The geometric relationship of the three-phase contact line corresponding to the droplet in states A and B was expressed as:

$$\pi L_B^2 - \pi L_A^2 = \lambda^2 \quad (4)$$

Similarly, the relationship between the surface energy barrier and the instantaneous wetting angle was described for the three-phase contact line transitioning from edge state C to adjacent edge state D in the double sinusoidal structure. The surface energy barrier from state C to state D was expressed as:

$$F_{C-D} = 2\pi \frac{L_D^2}{1 + \cos\theta_D} - \frac{L_C^2}{1 + \cos\theta_C} - 2\lambda \sqrt{h^2 + \frac{\lambda^2}{2}} \cos\theta_Y \quad (5)$$

The geometric constraint equation was shown as follows:

$$\frac{\pi L_C^2}{\sin\theta_C} \left(1 - \cos\theta_C - \frac{1}{3}1 - \cos\theta_C\right)^3 = \frac{\pi L_D^2}{\sin\theta_D} \left(1 - \cos\theta_D - \frac{1}{3}1 - \cos\theta_D\right)^3 + \frac{2}{3}\lambda^2 h \quad (6)$$

The geometric relationship of the three-phase contact line corresponding to the droplet in states C and D was expressed as:

$$\pi L_D^2 - \pi L_C^2 = \lambda^2 \quad (7)$$

The magnitude of the surface energy barrier represented the relative change in surface energy that the triple-phase contact line must overcome when moving from its initial state A to an adjacent state B (or from C to D). This barrier determined the ease with which the contact line can traverse a single repeating structural unit [31], directly controlling the ease of filler metal wetting and spreading.

3.3.2. Capillary driving force modeling

In contrast to the surface energy barrier generated by the vertical structure, the spreading of brazing droplet along parallel grooves is governed by capillary driving forces. To quantify the influence of parallel grooves and their amplitude on the wettability, a wetting kinetic model of filler metal within a single micro-groove was developed. The following assumptions were adopted to simplify the calculation: (1) the flow of brazing droplet within the groove is unidirectional; (2) the inertial effects are negligible; (3) except the threshold region, the liquid fills the groove to the top edge in the remaining region of the groove, and maintains a static advancing wetting angle θ with the sidewall (Fig. 5e), and the influence of local liquid morphology variations in the flow front (threshold region) on the total capillary driving force is neglected in this model; (4) the groove is sufficiently narrow to justify the assumption of a circular arc shape for the liquid surface. Emphatically, the spreading dwell time for molten AgCuTi was controlled to be relatively short prior to the initiation and thickening of the interfacial reaction layer. Consequently, the triple-phase line kinetics in the model were predominantly governed by surface tension and viscous resistance.

In a parallel micro-groove, the capillary force consistently acted to advance the liquid meniscus, independent of the local meniscus morphology. Thus, the capillary driving force [32] generated by the spontaneous spreading of the brazing droplet could be

described as:

$$F_{\sigma} = \gamma_{lv} P \cos \theta \quad (8)$$

where γ_{lv} was the surface tension of the filler metal at brazing temperature, P was the wetted perimeter of the groove, and θ was the advancing wetting angle. The capillary driving force had to overcome the viscous resistance encountered by the advancing liquid. The associated pressure drop over a distance z was described by the Darcy-Weisbach Eq. [33] as:

$$\Delta P = \frac{f_D \rho u^2 z}{2D_h} \quad (9)$$

where D_h was the hydraulic diameter of the groove, which was calculated as [33]:

$$D_h = 4 \frac{A}{P} = 2h \sin \beta \quad (10)$$

where the A represented the cross-sectional area of groove flow and P represented the wetted perimeter of the groove. β represented half of the apex angle of the groove. f_D represented the Darcy friction factor. ρ represented the density of the liquid filler metal. u represented the average flow velocity. Subsequently, the flow velocity of the liquid filler metal could be estimated using the Reynolds number formula [34]:

$$R_e = \frac{\rho u D_h}{\mu} \quad (11)$$

and the average flow velocity was expressed in terms of the interface propagation velocity $u = \frac{dz}{dt}$, allowing the viscous resistance to be written as:

$$F_{\mu} = \Delta P A = \frac{f_D R_e \mu P}{8D_h} z \frac{dz}{dt} \quad (12)$$

Equating the eqs. (9), (12) and integrating the result yielded the kinetic expression:

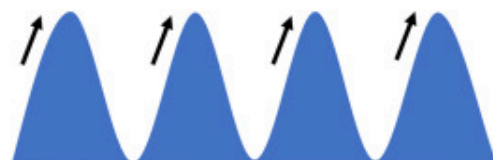
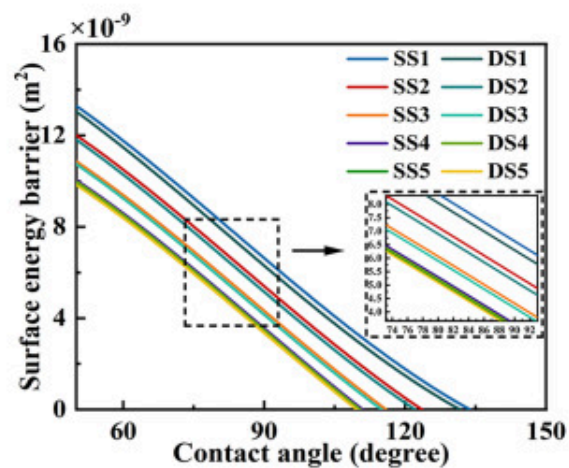
$$z^2 = \frac{32h\gamma_{lv}\cos\theta\sin\beta}{f_D R_e \mu} t \quad (13)$$

In the calculation of surface energy barrier, the parameters were set as follows: $L_A=L_C=1\text{ mm}$, $\theta_Y=110^\circ$. For the calculation of the capillary driving force, the advancing wetting angle θ was assigned the value of 85° in this study, which corresponded to the wetting angle of the brazing filler on the single sweeping surface. Owing to the lack of available data on the surface tension and viscosity of AgCuTi filler metal at brazing temperatures, this study adopted the corresponding values for AgCu alloy, which were given below: $\gamma_{lv}=0.957\text{ N/m}$, $\mu=0.0039\text{ Pa}\cdot\text{s}$. The Poiseuille number P_O , which is tabulated for the V-groove geometry by Shah and London (1978) [35], was used to determine the product $f_D R_e$ through the relation ($f_D R_e = 4P_O$).

3.3.3. Numerical results and discussion

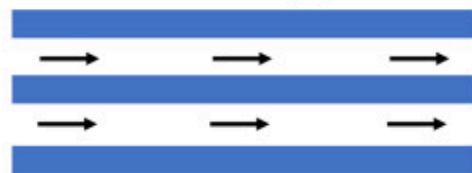
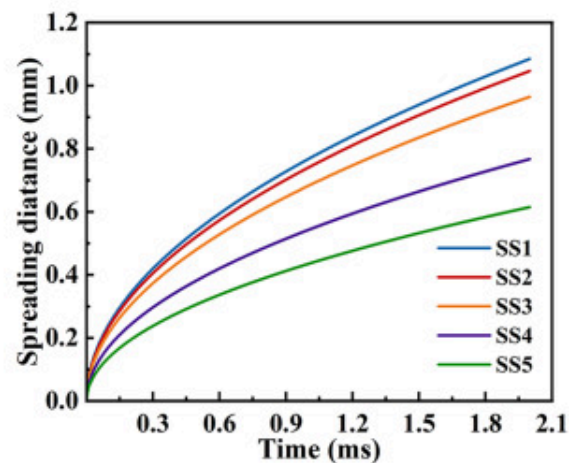
Based on the above eqs. (2–4) and (5–7), numerical calculation revealed the relationship between the variations in surface energy barriers and the instantaneous wetting angle θ for both single and double sinusoidal structured surface at different scanning speed groups, as shown in Fig. 6a. It could be observed that the surface energy barrier curves of the double sinusoidal structures at different amplitudes consistently lay below the single sinusoidal curves. This indicated that the energy barrier required for the triple-phase line to advance vertically traverse the micro-protrusions was lower than that needed to cross the micro-ridges of same projected length, as shown in the schematic diagram of Fig. 6a and c. Consequently, compared to the single sinusoidal structure, the triple-phase line moved forward more easily on the double sinusoidal structure surface. Furthermore, the difference in energy barriers between the double and single sinusoidal structures was more pronounced at lower scanning speed groups and gradually diminished as the scanning speed group increased. Meanwhile, the energy barriers of both single and double sinusoidal structure exhibited a decreasing trend with increasing scanning speed group, indicating a reduction in wetting resistance during the forward spreading of the triple-phase contact line. This mechanistically explained the overall decreasing trend of static wetting angles in the vertical direction as the scanning speed group increased. However, the slight increase in wetting angles simultaneously observed on the SS5 and DS5 surfaces was possibly related to the active surface effect.

(a)



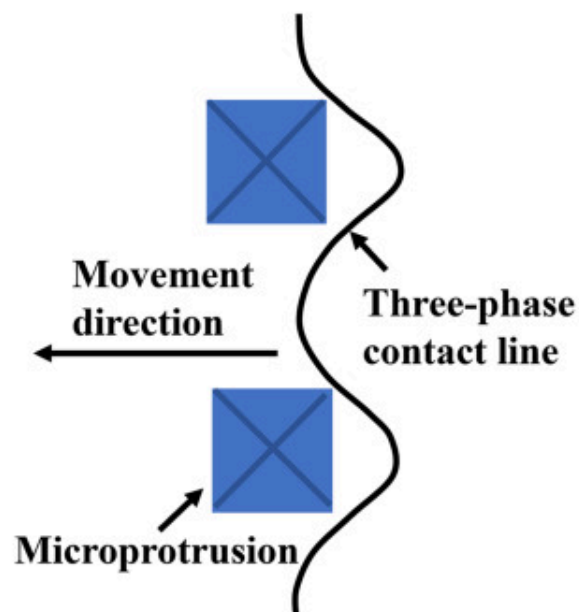
**Perpendicular
spreading direction**

(b)



**Parallel
spreading direction**

(c)



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Fig. 6. (a) Variation of the surface energy barrier with different amplitudes during perpendicular wetting process and single sinusoidal perpendicular spreading schematic, (b) capillary spreading distance over time during spreading along parallel grooves with different amplitudes and corresponding spreading schematic, (c) schematic diagram of the triple-phase contact line on the double sinusoidal structure.

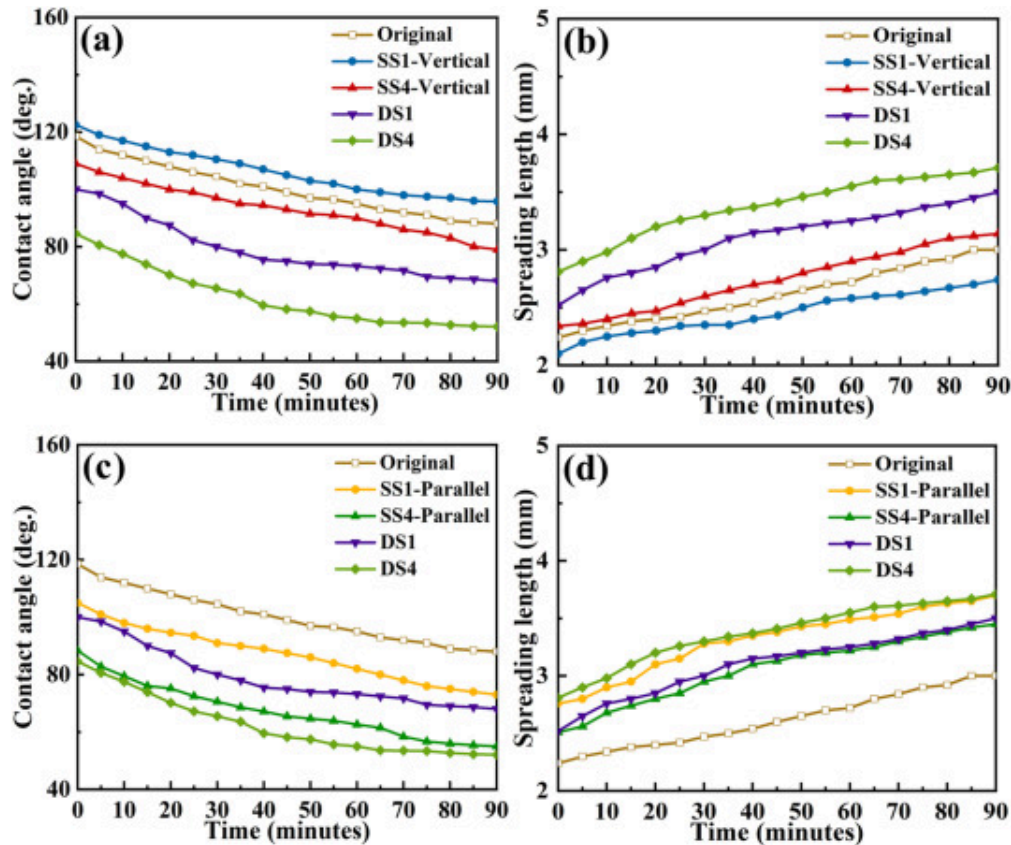
Conversely, when the triple-phase contact line advanced along the parallel direction of the single sinusoidal structure, its spreading was driven by capillary force without surface energy barrier, as shown in the schematic diagram of Fig. 6b. Numerical analysis based on Eq. (13) showed that the $z \sim t$ curves exhibited a consistent trend under different scanning speed groups (Fig. 6b). When the brazing droplet came into contact with the micro-groove, a rapid forward spreading of the liquid driven by capillary action was observed in the initial stage. As the triple-phase line advanced forward, the viscous resistance within the brazing liquid increased significantly with spreading velocity. To achieve instantaneous balance between the capillary force and the viscous resistance, the spreading velocity gradually decreased until the system eventually approached a quasi-steady state. This process demonstrated that the parallel groove enhanced the directional spreading capability of the filler metal by optimizing the dynamic balance between capillary force and viscous resistance. Further comparison revealed that higher scanning speeds, which produced shallower grooves, significantly weakened the capillary driving force. The underlying mechanism was that a decrease in groove depth reduces the wetted perimeter. From the perspective of energy, the reduction in wetted perimeter implied that less solid-gas interfacial area could be replaced per unit spreading distance. This directly diminished the amount of surface energy released per unit spreading distance ($F_{\sigma} = -\frac{dE}{dz}$), which was the fundamental reason for the reduction in the capillary force. However, this theoretically predicted trend was inconsistent with the experimental observations presented in Fig. 4, which showed that the wetting angles in the parallel

direction gradually decreased with increasing scanning speed. The discrepancy resulted from an interdependence between the wetting behaviors in the parallel and vertical directions: the pinning effect in the vertical direction restricted droplet spreading along that axis, thereby prompting increased accumulation of the brazing droplet along the edges of the parallel direction. This redistribution consequently increased the apparent wetting angles in the parallel direction, resulting in a consistent decreasing trend in wetting angles with increasing scanning speed in both directions. Therefore, the constraining effect of the capillary driving force (parallel direction) and the surface energy barrier (vertical direction) resulted in significantly different spreading capabilities of the filler metal, thereby leading to distinct anisotropic wetting behavior.

3.3.4. Dynamic wetting behavior

Fig. 7 comparatively analyzed dynamic wetting angles and spreading lengths of the filler metal along vertical and parallel directions on single and double sinusoidal structure surfaces, illustrating the influence of vertical surface energy barriers and parallel capillary forces on wettability. The timing started when the filler metal began to melt and formed a spherical shape. The initial wetting angle and spreading length of brazing droplet on the original SiO₂f/SiO₂ were 116° and 2.24mm, respectively. As the holding time increased, the stable wetting angle remained obtuse, and the corresponding spreading length changed less, confirming poor wettability. In comparison, when the droplet spread along the vertical structure direction on the single sinusoidal structure surface, the low scanning speed sample SS1 exhibited higher initial and the stable wetting angles and shorter spreading length compared to the original surface. As the scanning speed group increased to the SS4, the spreading rate surpassed that of both SS1 and the original surface, resulting in a lower stable wetting angle and a greater spreading length compared to the original surface. Furthermore, comparative analysis between single and double sinusoidal structures confirmed the influence of surface energy barriers on wettability. The initial wetting angles on double sinusoidal structure surfaces DS1 and DS4 were smaller than those on single sinusoidal structures SS1 and SS4, while the corresponding spreading lengths were larger. With the extension of time, significant

variations occurred in both the dynamic wetting angle and the spreading length. Consequently, the stable wetting angles followed the order: SS1-vertical > original surface > SS4-vertical > DS1 > DS4, while the corresponding spreading length exhibited the opposite trend.



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Fig. 7. Comparison of the dynamic wetting process of AgCuTi filler metal on double sinusoidal structures surfaces and single sinusoidal structures in the vertical and parallel directions under different scanning speed groups, respectively: (a) and (c) wetting angle variation, (b) and (d) spreading length variation.

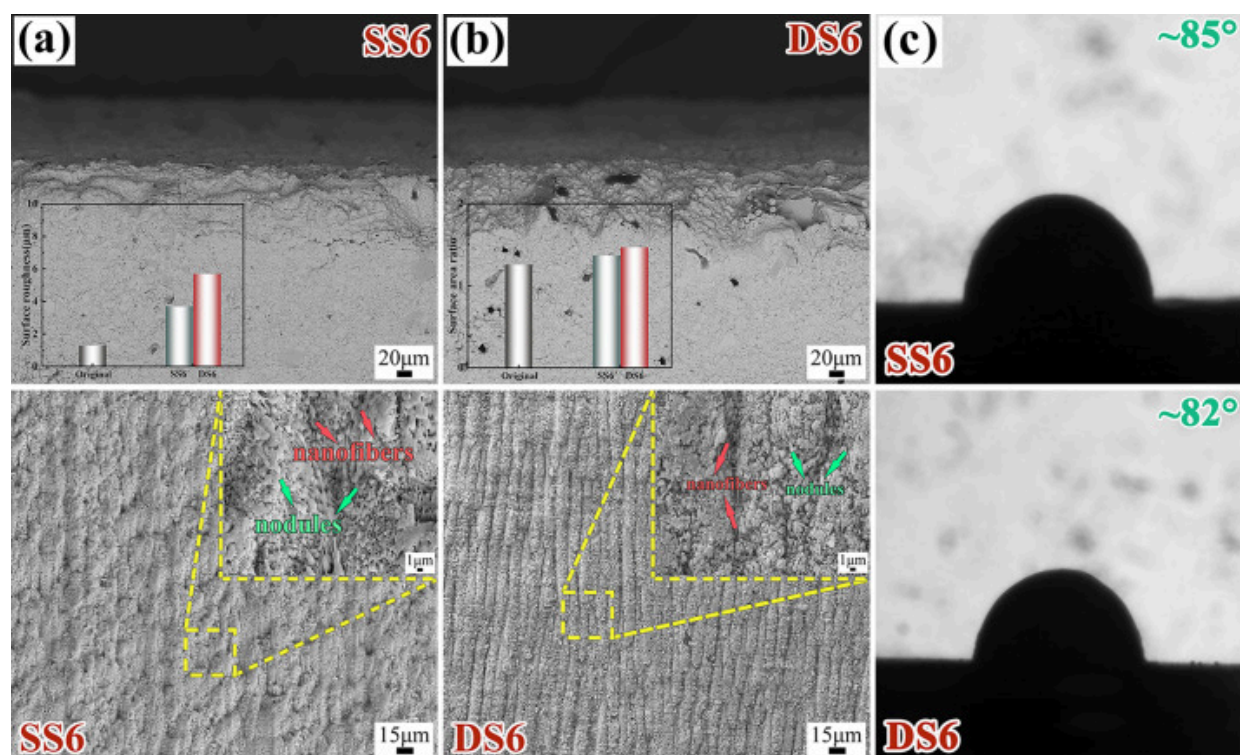
Conversely, when the droplet spread along the parallel structure direction under capillary driving force, both the SS1 and SS4 exhibited significant changes in spreading length, with their final spreading lengths exceeding that of the original surface. Furthermore, the change in the spreading length of SS1 was greater than that of SS4. However, influenced by the pinning effect in the vertical direction of the single sinusoidal structures, the stable wetting angles followed the trend original surface > SS1-Parallel > DS1 > SS4-Parallel > DS4 when compared with the double sinusoidal structures. And the spread length of DS4 was even slightly greater than that of SS1-Parallel, while the spread length of DS1 was marginally higher than that of SS4-Parallel. This indicated that the accelerated spreading rate of the brazing droplet on the double sinusoidal surface may have been influenced by both the structure morphology as well as the laser-induced active surface.

3.4. Influence of active surface on wettability

3.4.1. Sweeping surfaces preparation

Laser processing simultaneously alters both the macroscopic morphology and the surface state, creating a coupled effect on wettability. To disentangle the individual influences of macroscopic morphology and surface modification on wettability, sweeping samples were prepared using both single and double sinusoidal scanning strategies. This approach was designed to precisely eliminate the variable of macroscopic morphology while exclusively preserving the factor of laser-induced active surface. As shown in Fig. 8a-b, the cross-sections of the single and double sinusoidal sweeping samples exhibited no noticeable macroscopic morphology. The insets in Fig. 8a-b showed that the single and double sweeping surfaces possessed greater surface roughness and surface area ratios than the original surface. Analysis of their surface microstructure indicated that this could be attributed to the formation of a small number of elongated molten nanofibers and nodular features. The wetting angles of AgCuTi filler metal on the single and double sweeping surfaces were measured to be 85° and 82°, respectively, representing a significant decrease compared to the original surface (Fig. 8c). This reduction directly

confirmed the role of enhanced surface activity induced by laser in enhancing filler metal wettability.



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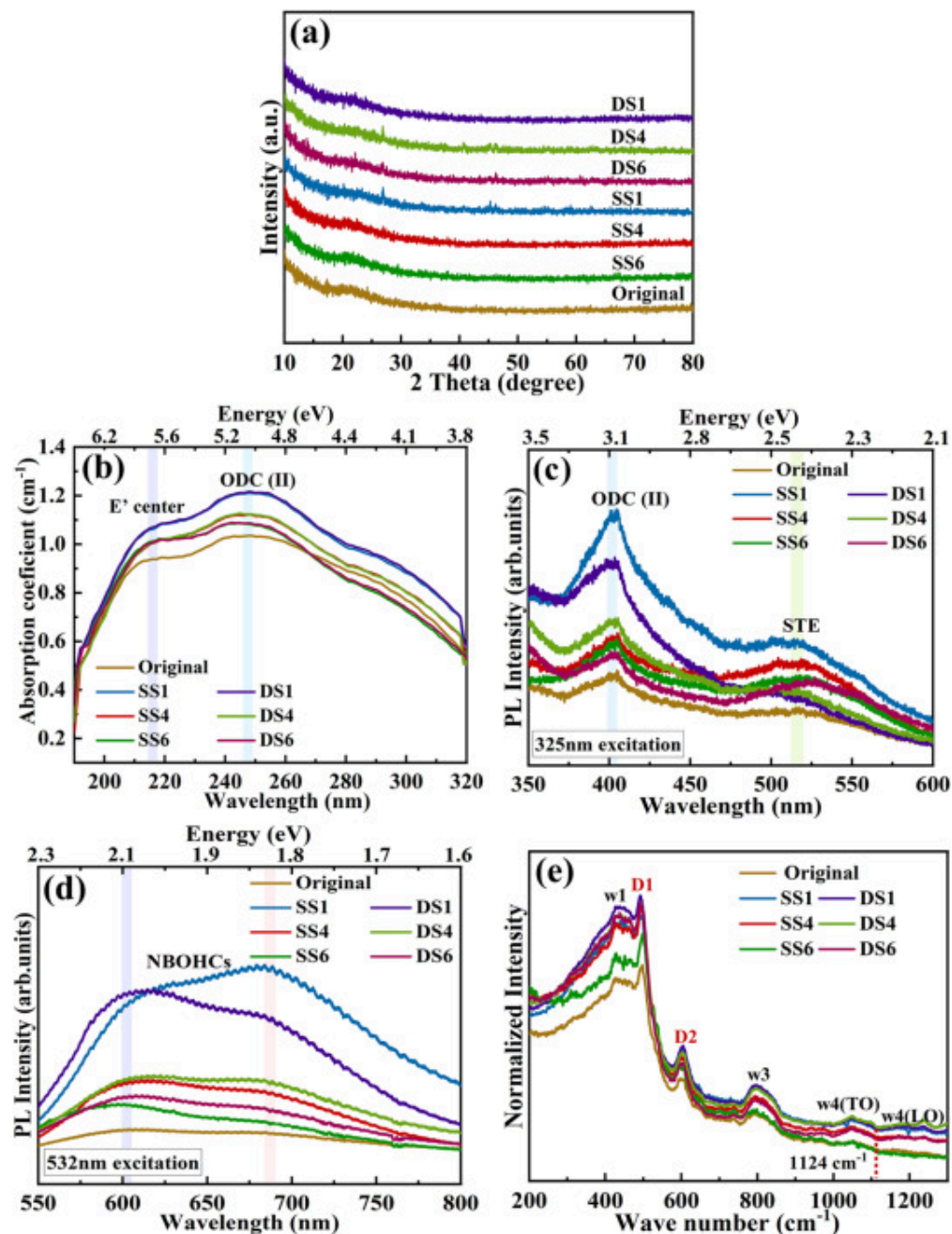
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Fig. 8. Single and double sinusoidal sweeping sample SS6 and DS6 at scanning speed group 900–1000mm/s: (a) SS6 and surface roughness values, (b) DS6 and surface area ratio values, (c) the wetting angles of the brazing filler metal on the sweeping samples.

3.4.2. Laser-induced active surface

The XRD patterns of the original surface and sinusoidal structured SiO₂f/SiO₂ surfaces under different scanning speed groups were initially analyzed to identify the phase transformations after laser processing. As shown in Fig. 9a, both the original surface and

the structured surfaces exhibited a “broad hump” peak, a typical feature of amorphous materials. This indicates that the high cooling rate of the picosecond laser prevented crystallization and formation of new phases.



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Fig. 9. (a) X-Ray diffraction patterns of the original surface, the single and double sinusoidal structures surfaces at different scanning speed groups, (b) ultraviolet-visible spectra of the original surface, the single and double sinusoidal structures surfaces at different scanning speed groups, (c-d) PL spectra of the original surface, the single and double sinusoidal structures surfaces at different scanning speed groups under excitation at 325 nm and 532 nm, respectively, (e) Raman spectra of original surface, the single and double sinusoidal structures surfaces at different scanning speed group.

Then, ultraviolet-visible (UV-vis) absorption, photoluminescence (PL), and Raman spectra were employed to characterize the laser-induced broken-bond defects and structural reorganization. Fig. 9b compared the variation in the UV-vis absorption coefficient between the original surface and sinusoidal structured surfaces. The spectra showed two distinct absorption bands at 214 nm and 248 nm. The 214 nm (5.8 eV) peaks were attributed to the E' center defects ($O_3 \equiv Si \cdot$), which involve an unpaired electron on a silicon atom bonded to three oxygen atoms [36]. The peaks centered at 248 nm (5 eV) corresponded to the ODC (II) point defects, which are related to the loss of two oxygen atoms in the lattice and can be represented as ($O_2 = Si:$) [37]. The absorption coefficient intensity slightly decreased with increasing laser scanning speed group. Additionally, low density defects were detected on the original surface. The spectrometer's lower wavelength limit of 190 nm prevents the direct detection of reported defects (ODC(I), ODC(II), and NBOHC) absorbing below this range [38]. Therefore, (PL) spectra were employed for their characterization.

Fig. 9c presented the PL spectra obtained under excitation at 325 nm for original surface and sinusoidal structured surfaces. The spectra featured relatively sharp peaks at 400 nm and broad peaks around 530 nm, which were attributed to the oxygen deficiency center ODC (II) defects (emission at ~3.1 eV) [39] and self-trapped exciton (STE) defects (emission at ~2.3 eV) [40], respectively. The STEs were defined as localized electron-hole pairs in a lattice distortion, represented by $O_3 \equiv Si \cdots O - Si \equiv O_3$ [40]. Additionally, corresponding spectra under 532 nm excitation for both surfaces were presented in Fig. 9d. According to the literature [41], the emissions in the range of 620 nm-680 nm were

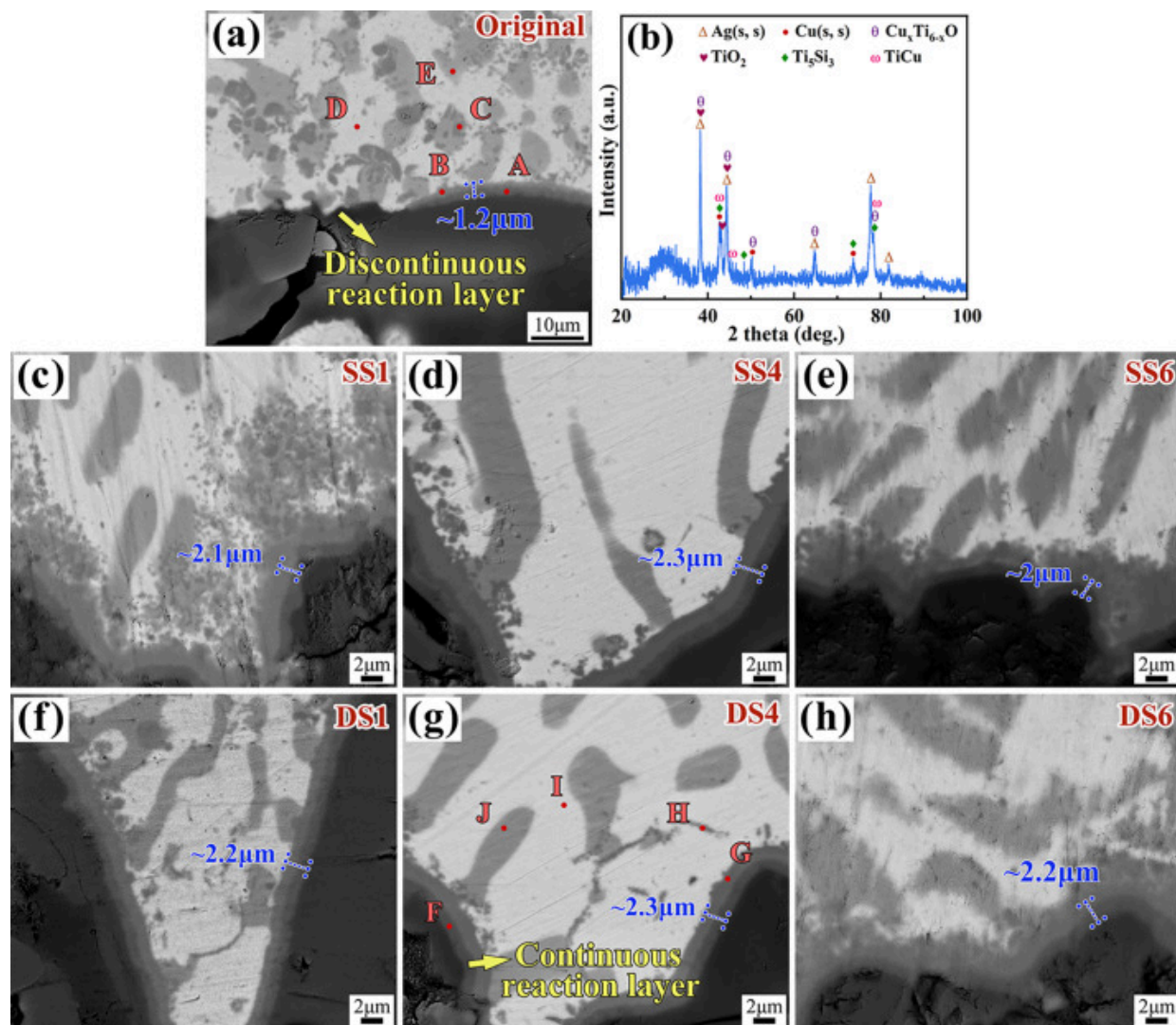
attributed to the defects NBOHC, defined as non-bridging oxygen atoms that have lost electrons ($O_3 \equiv Si - O$). Therefore, laser processing generated a high density of active defects on the sinusoidal structured surface without changing the surface chemical composition. Moreover, the PL spectra revealed a consistent trend for both sinusoidal structured surfaces, the defect concentration gradually decreased with increasing scanning speed group but remained higher than that of the original surface. The intrinsic defects on the original surface had been confirmed to enhance the absorption of laser energy [42].

The formation of defects inevitably disrupts the network structure of SiO₂f/SiO₂ composite, making it essential to characterize the microstructural changes. Micro-Raman spectroscopy has been proven to be an effective method for characterizing structural modifications. The normalized typical Raman spectra of the original and structured surfaces were shown in Fig. 9e. The main bands w_1 at 445cm^{-1} were attributed to the symmetric stretching vibration of the Si – O – Si bond angle characteristic in the high-membered rings [43]. The two relatively sharp peaks centered at 490cm^{-1} and 605cm^{-1} were assigned to the in-phase breathing motions of oxygen atoms in folded four- and planar three-membered ring structures, known as defect D_1 and D_2 , respectively [44]. The w_3 peaks located at 805cm^{-1} were attributed to the local bending and stretching of the Si – O – Si bonds [45]. The longitudinal and transverse optical (TO-LO) doublet w_4 centered at 1124cm^{-1} was related to the asymmetric stretching of the Si – O – Si bonds [43]. Comparison revealed that the relative intensities of the defect peaks D_1 and D_2 in the sinusoidal structured surfaces increased compared to the original surface. This means that partial Si – O bonds would break in some of six- and five-membered rings, leading to their transformation into unstable small-membered rings [44]. Moreover, the number of four- and three-membered rings declined as the scanning speed groups increased. Based on the preceding analysis, the laser-induced defects exhibited a gradually decreasing trend with increasing scanning speed. The surface defect density was governed by the effective energy input, as quantified by Eq. (1). Specifically, increasing the scanning speed reduced the accumulated energy per unit area, leading to a

systematic decrease in defect density. In comparison, the single and double sweeping samples were prepared using a single-scan strategy, resulting in lower accumulated energy than the sinusoidal structured samples. Consequently, the observed defect density of these sweeping samples was lower than their sinusoidal samples but remains higher than that of the original surface.

3.4.3. Wetting interface microstructure

To investigate the influence of laser-induced defects on the interfacial reaction, the interfacial microstructure of AgCuTi filler metal on original and sinusoidal structured surfaces at different scanning speed groups was compared in Fig. 10. The original interface exhibited a discontinuous reaction layer measuring only 1.2 μm in thickness, indicating limited interfacial reactivity (Fig. 10a). In contrast, sinusoidal structured interfaces consistently displayed continuous interfacial reaction layers with thicknesses of 2–2.3 μm (Fig. 10c-h). This suggested that the laser-induced defects enhanced the interfacial reaction activity, thereby promoting the metallurgical bonding. A detailed microstructure analysis was conducted on the brazing seam and adjacent reaction layers of the DS4 and original interfaces (Fig. 10a and g). And the chemical composition at the marked points (Table 1) as well as XRD pattern (Fig. 10b) of the seam were characterized. Based on combined XRD and EDS results, the microstructure of the brazing seam mainly consisted of Ag-based solid solution and Cu-based solid solution. The gray small particles near the interfacial reaction layer (points C and H) were identified as Ti₃Cu₄ compounds, which are dispersed within the Ag—Cu solid solution. At the SiO₂f/SiO₂ interface, EDS analysis of points A-B and F-G confirm the formation of a bilayer reaction zone, which consisted of the Ti₅Si₃ and TiO₂ phases adjacent to SiO₂f/SiO₂ composite and the Cu₃Ti₃O compounds near the brazing seam. Therefore, compared to the original interface, the sinusoidal structure interface did not alter the phase constituents, it predominantly modified the interfacial morphology and enhanced the continuity and thickness of the reaction layer. Moreover, neither the composition nor the thickness of the reaction layer exhibited significant changes under different laser parameters.



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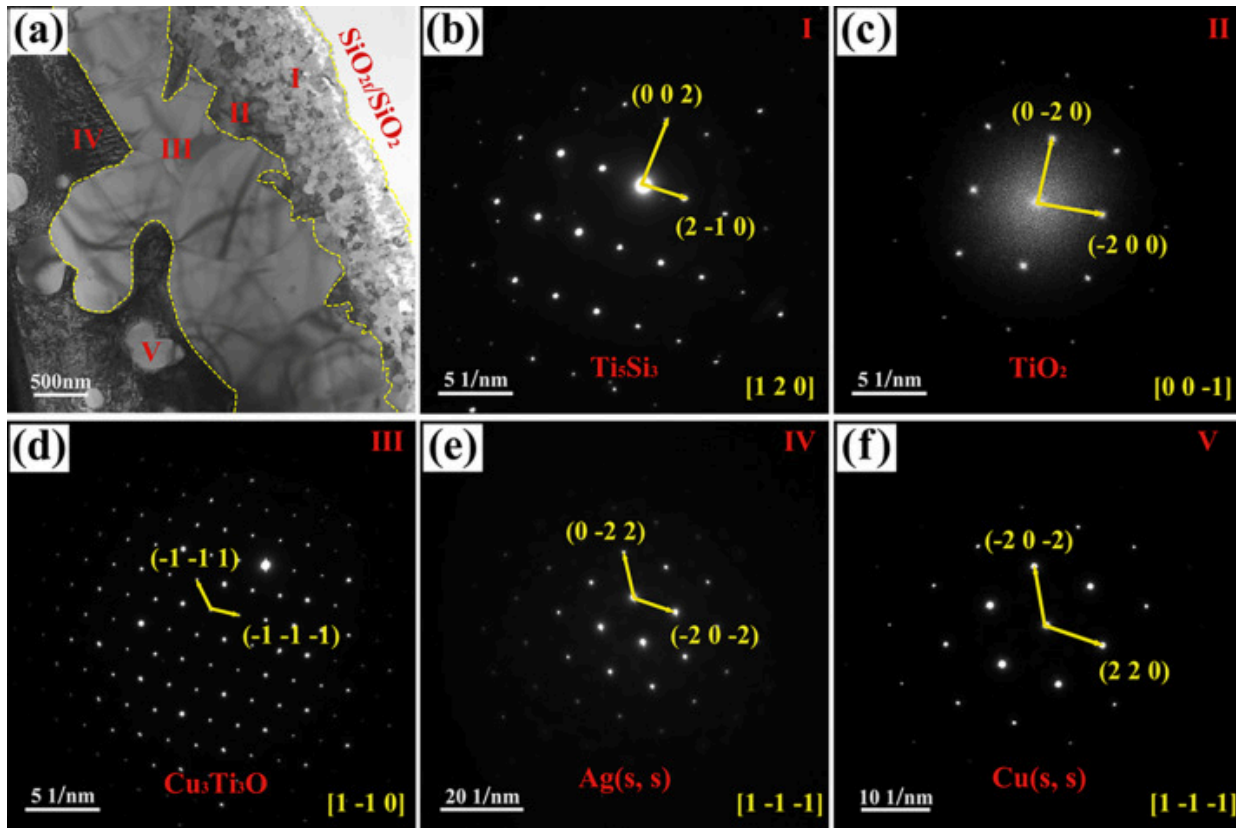
Fig. 10. Interfacial microstructure of the AgCuTi-SiO₂f/SiO₂ wetting system with different interfaces: (a) interface on original SiO₂f/SiO₂, (b) XRD pattern of the brazed seam with DS4 processed surface, interfaces on structured SiO₂f/SiO₂ with different scanning speeds: (c) SS1, (d) SS4, (e) SS6, (f) DS1, (g) DS4, (h) DS6.

Table 1. Average chemical compositions of each reactant (at.%) in enlarged view of Fig. 10 a and g.

	O	Al	Si	Ti	V	Cu	Ag	Possible phases
A	12.89	1.36	32.39	48.78	0.05	4.36	0.17	Ti-O and Ti-Si
B	5.63	8.38	5.72	45.9	0.51	33.56	0.30	Ti-Cu-O
C	–	6.30	1.24	37.97	0.39	45.65	8.45	Ti ₃ Cu ₄
D	11.72	0.74	–	0.47	0.19	13.92	72.97	Ag(s, s)
E	3.75	4.38	–	1.39	0.09	88.79	1.60	Cu(s, s)
F	14.68	1.68	32.33	45.66	0.41	5.10	0.14	Ti-O and Ti-Si
G	12.91	8.01	6.67	42.25	0.26	29.39	0.51	Ti-Cu-O
H	7.52	6.69	6.48	34.40	–	41.64	3.28	Ti ₃ Cu ₄
I	12.69	1.19	–	0.95	–	12.94	72.24	Ag(s, s)
J	2.47	4.58	–	1.35	0.04	89.58	1.98	Cu(s, s)

To accurately determine the structure and composition of the interfacial reaction layer, further characterization by TEM was required. Fig. 11a shows the bright field TEM image of the reaction layers adjacent to the structured (DS4) SiO₂f/SiO₂ side. Evidently, the brazing seam consists of Ag—Cu solid solution phase. The double reaction layers consisted of a thin layer (approximately 1 μm thick) adjacent to the SiO₂f/SiO₂ composite and the thicker layer (approximately 1.5 μm thick) on the brazing filler side. Based on the corresponding selected area electron diffraction (SAED) analysis (Fig. 11b-c), the thin layer was composed of phase I identified as Ti₅Si₃ with the hexagonal structure, and phase II confirmed to be TiO₂ with the tetragonal structure. Meanwhile, the thicker layer (phase III) was identified as the Cu₃Ti₃O phase with the cubic structure, as confirmed by the SAED pattern presented in Fig. 11d. Notably, the Cu₃Ti₃O phase, which could be

classified as M_6X type compound, exhibited metal-like characteristics due to its electrical resistivity ($\rho=5\times 10^{-6}\Omega\cdot m$) being close to that of the Ti alloy ($1.5\times 10^{-6}\Omega\cdot m$) [46]. This property ensures good compatibility with the molten AgCuTi filler metal, facilitating its spreading over the reaction layer, which is the key reason why the formation of Cu_3Ti_3O compound promotes wetting in this system [47]. Based on the previous analysis, an increase in scanning speed group resulted in a reduced surface area and a lower defect density of the sinusoidal structure, which indicated a decrease in active area. This reduction was unfavorable for the rapid formation of Cu_3Ti_3O compounds, potentially explaining the slight increase in the wetting angles observed for SS5 and DS5. Therefore, the active defects on the sinusoidal structured surface significantly facilitated the rapid formation of a continuous and thickened reaction layer. This layer exhibited distinct functional gradient characteristics: the outer Ti_5Si_3/TiO_2 compound layer formed a robust metallurgical bond with the ceramic matrix. The inner Cu_3Ti_3O layer exhibited excellent compatibility with the AgCuTi filler metal, thereby effectively enhancing its wettability and spreading.



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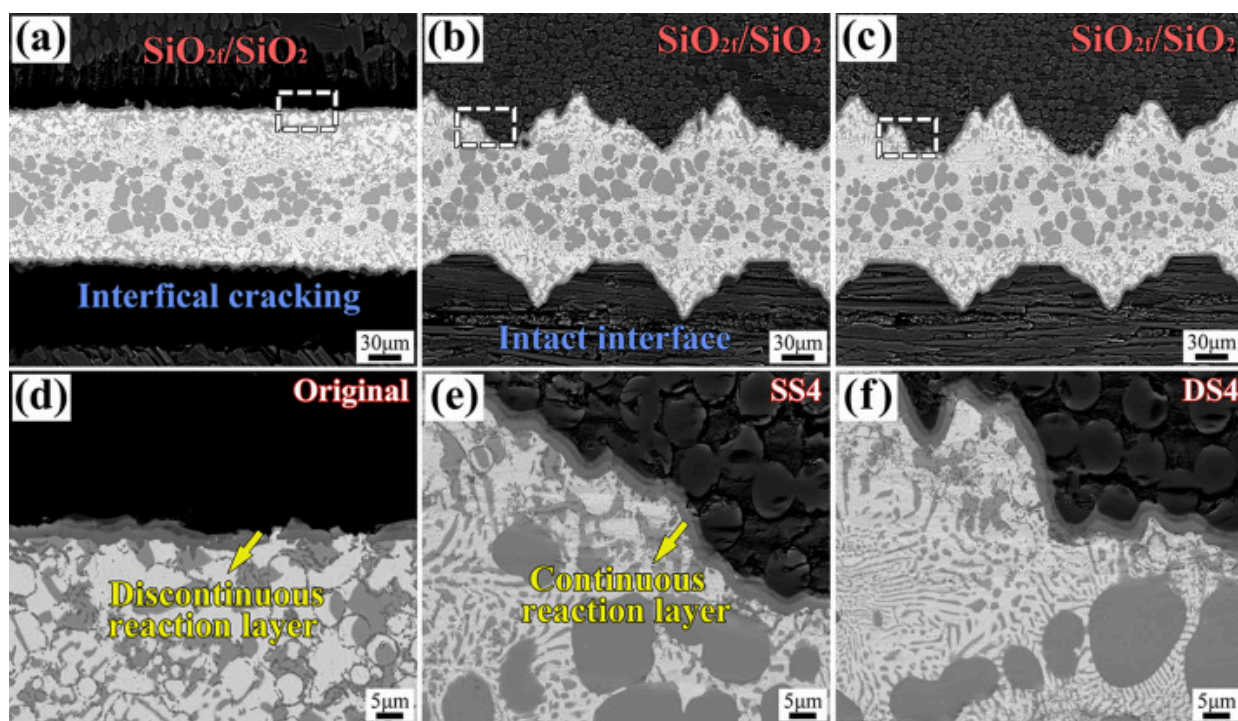
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Fig. 11. TEM images of reaction layer on DS4 processed SiO₂f/SiO₂ composite side interface and corresponding selected area electron diffraction (SAED).

3.5. Interfacial morphology and strength of SiO₂f/SiO₂ joints

With the wetting mechanism established, the interfacial morphology and mechanical strength were subsequently assessed to validate the beneficial role of sinusoidal structures in joint quality. Fig. 12 shows the typical cross-sectional microstructure of joints with three distinct interfacial bonding modes: the original interface, the bioinspired single (SS4) and double (DS4) sinusoidal structured interfaces. In the original

joint (Fig. 12a), notable cracking was observed in the SiO₂f/SiO₂ composite adjacent to the brazing interface, originating from residual stresses induced by CTE mismatch. Meanwhile, due to the strong chemical inertness of the SiO₂f/SiO₂ composite, the interfacial reaction was inadequate. The resulting reaction layer was incapable of withstanding the stresses, thus compromising its integrity and becoming discontinuous (Fig. 12d). In comparison, the bioinspired sinusoidal structured joints demonstrated excellent bonding integrity without defects, as shown in Fig. 12b-c. Notably, the sinusoidal structures were fully infiltrated by the brazing filler, forming continuous and dense reaction layers along the alternating peak-valley topography at the composite/brazing filler interface.



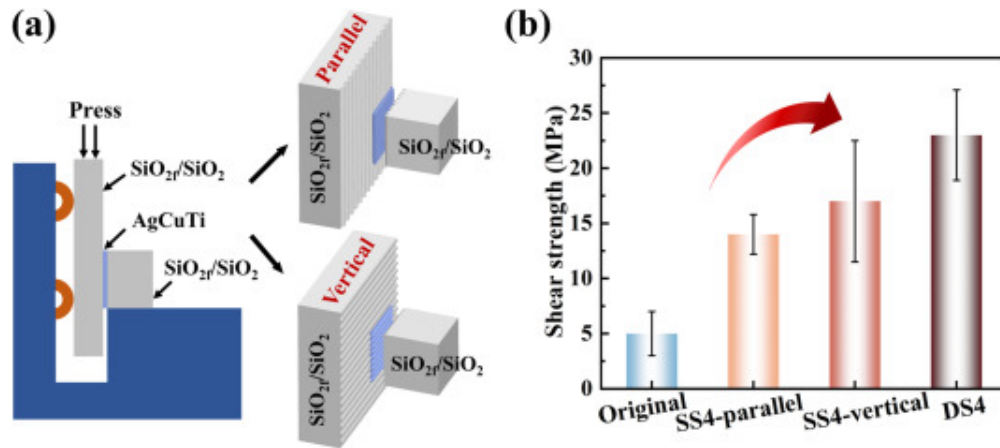
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Fig. 12. Interfacial microstructure comparison of SiO₂f/SiO₂/AgCuTi/SiO₂f/SiO₂ joints with three interface states, (a) original, (b) single sinusoidal structure SS4, (c) double sinusoidal

structure DS4, (d-f) partial magnification of the interfacial reaction layer corresponding to (a-c), respectively.

The microstructure of the brazed joints directly determines the mechanical properties of the joints. To this end, the effect of the bioinspired sinusoidal structure on the shear strength of the brazed joint was investigated (Fig. 13b). The original SiO₂f/SiO₂ joint exhibited remarkably low shear strength, averaging only 5 MPa. Accounting for the anisotropy of the single sinusoidal structure, shear loading was conducted to the brazed joints independently along both the vertical and parallel structural directions (Fig. 13a). The shear strength parallel to the structural direction (~14 MPa) was lower than that in the vertical direction (~17 MPa), but still greater than the original interface joint strength. Moreover, the shear strength of the double sinusoidal structured joint (23 MPa) not only surpassed that of the single sinusoidal structured joint along the vertical direction but also reached 4.6 times that of the original interface joint. This significant improvement in joint strength was associated with the stress mitigation facilitated by the sinusoidal structure. The stress mitigation mechanism of the sinusoidal structure was consistent with that reported in the literature [[48], [49], [50]]. Firstly, the extended interfacial path reduced stress value per unit length, which mitigates localized stress concentration and suppresses crack initiation. Secondly, the periodic sinusoidal profile induced mechanical interlocking and forced propagating cracks to deflect along a tortuous path, thereby increasing the energy required for crack propagation and enhancing the overall fracture toughness of the joint.



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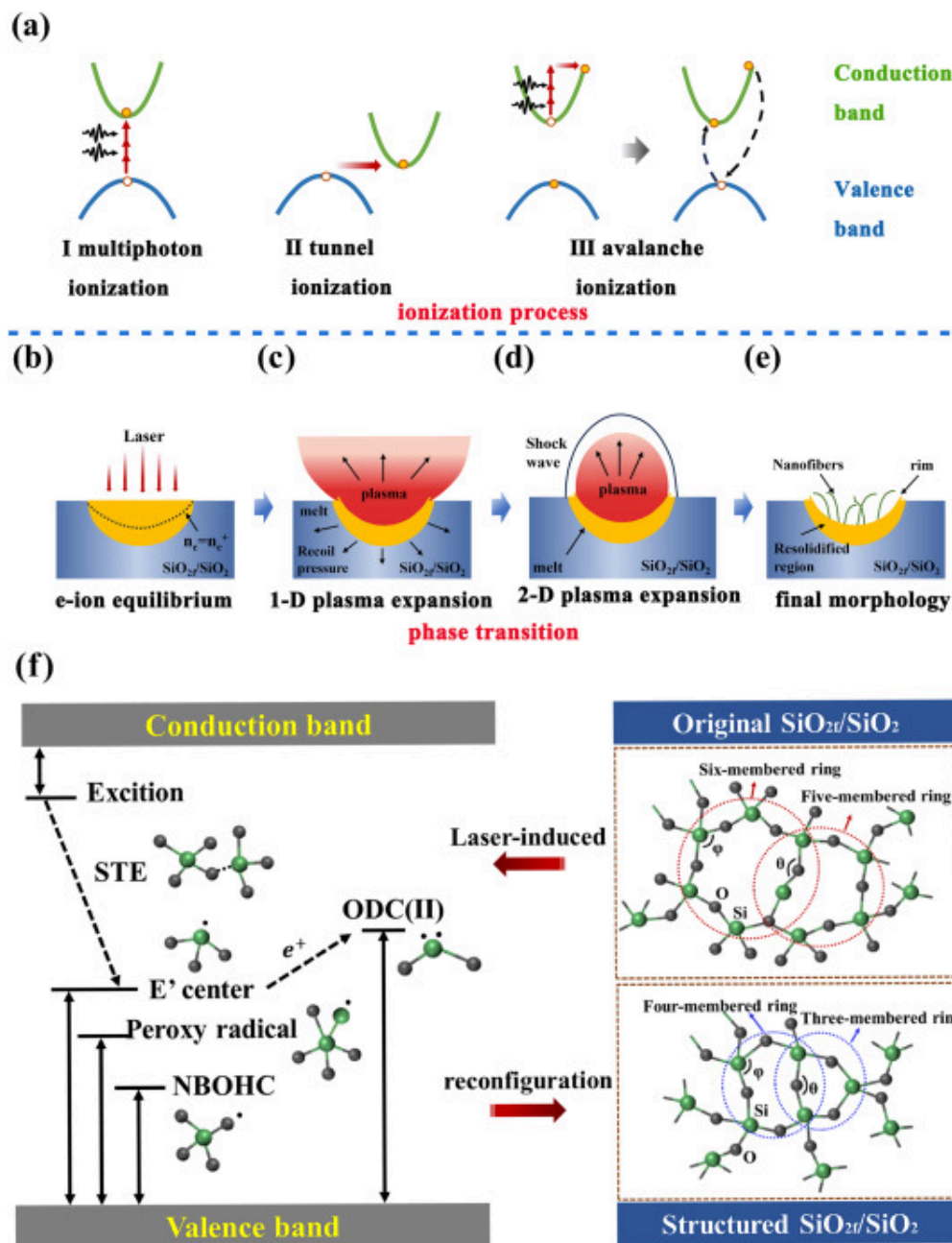
Fig. 13. (a) schematic structure of the shear fixture and testing on the single sinusoidal structure in both vertical and parallel directions, (b) comparison of shear strength among brazing joints with three interface states.

4. Discussion

4.1. Laser-SiO₂f/SiO₂ interaction mechanism

The ablation process of the wide bandgap ($\sim 9\text{eV}$) SiO₂f/SiO₂ composite under laser pulses consists of two stages: ionization and phase transition, as depicted in Fig. 14. Irradiation of SiO₂f/SiO₂ composite with 325nm (3.81 eV) picosecond laser pulses provides insufficient photon energy for valence-to-conduction band excitation. Consequently, valence electrons must achieve transition to the conduction band by simultaneously absorbing multiple photons, ionizing into free electrons and positive ions [51]. This process known as multiphoton ionization, is illustrated in part I of Fig. 14a. Tunnel ionization is another critical mechanism in non-metallic materials, dominated by the intense laser field. The superposition of the nucleus Coulomb field and the laser field

creates an oscillating finite potential barrier, allowing valence electrons to tunnel into the conduction band via quantum tunneling [52], as shown in part II of Fig. 14a. These free electrons are promoted to higher energy states by absorbing laser photons. Once their energy exceeds the bandgap of the conduction band, they can ionize another electron from the valence band through impact ionization, resulting in two electrons at the conduction band minimum. This cycle of photon absorption and impact ionization repeats, thereby initiating an avalanche (the part III of Fig. 14a) [53].



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Fig. 14. Schematic description of ablation process of SiO₂f/SiO₂ with picosecond laser pulse (a) ionization process, (b) e-ion equilibrium, (c) 1-D plasma expansion, (d) 2-D plasma expansion, (e) final morphology, (f) laser-induced formation of surface defects and the microstructural network reorganization process on SiO₂f/SiO₂ surface.

Through the ionization mechanism, a large number of free electrons and positive ions accumulate within the material, subsequently entering the phase transition stage. The transfer of absorbed laser energy from free electrons to the lattice occurs through electron-phonon scattering, a relatively slow process that continues after the laser pulse ends. As the electrons and ions thermal equilibrate through collisions, a high-temperature and high-pressure plasma forms above the sample surface (Fig. 14b) [54]. The plasma generated by the ultrafast laser reaches pressures of several million atmospheres within tens of picoseconds, and initially expands vertically from the material surface (Fig. 14c). This pressure then decreases to several tens of atmospheres within tens of nanoseconds, starting to expand radially. The expanding plasma releases most of the absorbed energy into the surrounding gas, while a small amount of thermal energy remains on the material surface. When multiple pulses strike the same focal area at a repetitive frequency, heat accumulation becomes profound as the time interval between the pulses is shorter than the time required for the absorbed heat to diffuse out of the focal volume. The accumulation of heat elevates the temperature within the focal volume to approximately 3000°C, thereby inducing material melting and the molten pool formation beneath the expanding plasma [55] (Fig. 14d). The movement of molten material is primarily driven by thermal capillary forces and plasma recoil pressure. However, for the SiO₂f/SiO₂ materials, the effect of the positive surface tension gradient on driving thermal capillary flow is negligible due to the high melt viscosity [54]. Driven by plasma recoil pressure, the molten material flows from the center to the edge of the groove. When heat dissipation causes the temperature to drop below the melting point, re-solidification occurs at the edge of the groove, forming a raised rim [54]. As the hot melt undergoes rapid cooling at the edge, the forefront ejects dense nanofibers, while the rear contracts into small spheres that adhere to the material surface [55], as shown in Fig.14e. Following

laser irradiation, the single and double sinusoidal structural characteristics were formed, as shown in Fig. 2 and Fig. 8a-b.

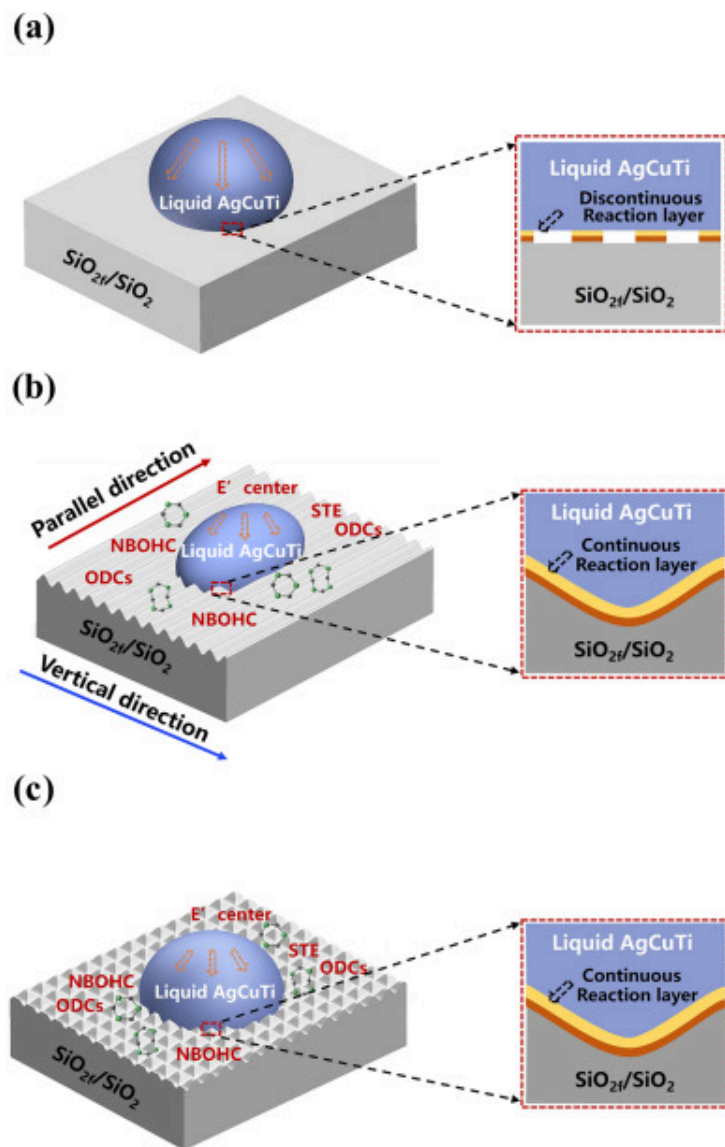
Laser energy deposition through nonlinear absorption not only alters the morphology of SiO₂f/SiO₂ composite but also permanently modifies its random network structure, which is composed of fundamental SiO₄⁴⁻ tetrahedral units [56]. This structure features silicon atoms bonded to two oxygen atoms, forming six distinct closed-ring types. The majority are five-membered or six-membered rings, where the O – Si – O bond angle is 109° due to the minimal strain energy required for their formation (Fig. 14) [57]. In contrast, forming three- or four-membered rings requires a smaller O – Si – O bond angle of 102°, resulting in distortion and instability of the tetrahedral structure [57]. Furthermore, laser irradiation generates a series of bond-breaking defects, resulting in a highly active surface, as illustrated in Fig. 14f. According to the previous discussion, multiphoton absorption in the composite generates holes in the valence band and electrons in the conduction band, which couple to form excitons [58]. Subsequently, strong electron-phonon coupling induces lattice relaxation and distortion, leading to the formation of self-trapped exciton (STE) defects [58]. Under continued laser irradiation, a portion of the STE defects undergo Si ... O bond breaking, resulting in the creation of E' centers accompanied by the formation of peroxy radicals (Si – O – O ·). Concurrently, other STEs may break terminal Si – O bonds, forming nonbridging oxygen vacancy centers (NBOHC) [59]. Subsequent oxygen-deficient centers ODC (II) are formed from E' centers through a process of electron capture, charge neutralization, and lattice relaxation [39].

The formation of defects disrupts the network structure of SiO₂f/SiO₂, inducing microscopic lattice reorganization. As evidenced by the Raman spectroscopy in Fig. 11e, the reorganization results in a significant increase in the intensity of three- and four-membered ring peaks, indicating a decrease in bridging angles and a densification of the laser-modified region compared to the original surface. Marcinkevičius et al. found that in the densified silica, the decrease in bridging angles due to the deformed configuration of the oxygen's valence electrons increases the reactivity of oxygen [60]. Therefore, the

presence of defects and an increased population of three- and four-membered rings within the micro-grooves and micro-pits enhances the interfacial metallurgical reactions.

4.2. Wetting mechanism

Based on the preceding analysis, the wetting mechanism of AgCuTi filler metal on the original, single and double sinusoidal surfaces is systematically elucidated, with the corresponding model presented in [Fig. 15](#). The high chemical inertness of the original SiO₂/SiO₂ surface results in relatively slow interfacial reactions, which leads to a discontinuous Ti₅Si₃/TiO₂/Cu₃Ti₃O reaction layer. This discontinuity, particularly in the metallic Cu₃Ti₃O phase, hinders the wetting and spreading of the filler metal. Furthermore, the inadequate metallurgical bonding weakens the interface, thereby making the joint susceptible to cracking from stress mismatch during cooling, as shown in [Fig. 15a](#).



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Fig. 15. Diagram of the wetting and spreading mechanism for the AgCuTi-SiO₂f/SiO₂ system with three different surfaces: (a) original surface, (b) single sinusoidal structure, (c) double sinusoidal structure.

Conversely, the spreading perpendicular to the groove of the single sinusoidal structure, the micro-ridge structures induce significant surface energy barriers that oppose the advancing triple-phase line (Fig. 15b). This resistance results in pinning at the structural edges, thereby restricting the lateral spreading of the molten filler metal. In this direction, the wettable Cu₃Ti₃O reaction layer governs the spreading behavior by replacing the high-interface energy filler/SiO_{2f}/SiO₂ interface with a low-interface energy filler/Cu₃Ti₃O interface. The resultant local reduction in the surface energy barrier at the micro-ridge tops creates an energy pathway, enabling the triple line to climb over the micro-ridges. However, the energy contribution from the interfacial reaction is localized, occurring only in the immediate vicinity of the triple-phase line. Consequently, it is incapable of altering the overall geometric energy barrier established by the micro-ridge array. Under this constraint, the triple line ultimately stabilizes in the energy-minimum groove, forming a new pinning point. This wettability arises from the competition between localized chemical driving forces and geometric constraints. As the scanning speed increases, the surface energy barrier is reduced, allowing the chemical driving force to become operative, thereby leading to enhanced wettability.

During the wetting and spreading along the parallel direction of the single sinusoidal structure are governed by a synergistic interaction between surface tension-induced capillary flow and interfacial chemical reactions (Fig. 15b). The capillary forces generated by the parallel groove structure drive the molten filler metal to rapidly infiltrate the grooves, significantly increasing the actual contact area with the SiO_{2f}/SiO₂ composite and thereby providing an expanded interface for the reaction. Simultaneously, the laser-induced Si—O broken bonds and metastable small-membered ring defects serve as active sites, substantially promoting the interfacial reaction between the active element Ti and SiO_{2f}/SiO₂ composite. This facilitates the rapid formation of a wettable Cu₃Ti₃O reaction layer, which reduces the local interfacial energy and decreases the wetting angle. The resultant enhancement in capillary driving force propels the triple-phase line forward, ultimately establishing a self-promoting wetting process. However, the wetting angles

parallel to the grooves also gradually decrease with increasing scanning speed group, a trend influenced by the pinning effect originating from the perpendicular direction.

The double sinusoidal structure exhibits superior wettability to the single sinusoidal structure in both parallel and vertical directions, a performance advantage attributed to the synergistic effect of two mechanisms (Fig. 15c). From the perspective of macroscopic morphology, the orthogonal processing method decomposes the continuous micro-ridges into uniformly distributed micro-protrusion arrays. Energy barrier calculations reveal a substantially lower surface energy barrier for the molten filler metal to traverse a single micro-protrusion relative to a continuous micro-ridge of equivalent projected length. This reduction in the energy barrier effectively suppresses the pinning of the triple-phase contact line and thereby greatly facilitating the spreading process. Secondly, the double sinusoidal design simultaneously constructs an interconnected and isotropic micro-grooves network, providing abundant preferential capillary channels for the filler metal to spread in any in-plane direction. Furthermore, it also possesses a significantly larger highly active surface area compared to the single sinusoidal structure, promoting the rapid formation of the wettable Cu₃Ti₃O reaction layer, which further accelerates the spreading process. In summary, the synergy among the reduced energy barrier, multi-directional capillary driving force, and the enhanced interfacial reaction collectively drives the filler alloy to achieve rapid and uniform omnidirectional spreading, ultimately resulting in isotropic wetting performance significantly superior to that of the single sinusoidal structure.

As indicated by the preceding analysis, the double sinusoidal structure exhibited optimal wetting behavior at scanning speeds of 900–1000 mm/s (DS4), with the wetting angle decreasing to approximately 70°. The corresponding SiO₂f/SiO₂ joint strength increased to 23 MPa, reaching 4.6 times that of the original joint, providing direct evidence that wettability improvement contributes significantly to joint reliability. The improved wettability enabled by the sinusoidal structure could be attributed to its facilitation of thorough interfacial spreading and capillary penetration of the filler metal.

Simultaneously, laser-induced active surface provided preferential nucleation sites for the

reaction phases. These combined effects promoted the formation of a continuous and dense interfacial reaction layer, effectively suppressing joint defects. Furthermore, the continuous interfacial layer enhances the load-bearing capacity of the joint, collectively contributing to the improvement of its mechanical properties.

5. Conclusions

This study investigated the high-temperature wetting behavior mechanism through the synergistic effect of bioinspired sinusoidal structure morphology and laser-induced active surfaces. The effectiveness of this strategy in addressing ceramic joining challenges was demonstrated by the enhanced wettability and joint strength in SiO₂f/SiO₂ composites joints.

However, this study has limitations in terms of processing depth control. Excessive depth can introduce microcracks and thermal damage, thereby compromising the mechanical properties of the ceramics. A promising future direction involves fabricating secondary sinusoidal features based on the primary sinusoidal structure. This synergistic strategy is expected to further improve wettability by increasing the active defect area while enhancing joint strength through effective stress dispersion. The conclusions of this study are as follows:

- (1) Bioinspired single and double sinusoidal structures were manufactured on the SiO₂f/SiO₂ surfaces. Numerical modeling revealed that the macroscopic morphology governed the spreading behavior of the brazing droplet. When moving vertically across the micro-ridges on the single sinusoidal structure surface, it had to overcome the surface energy barrier. In contrast, capillary forces within the micro-grooves significantly promoted spreading in the parallel direction. By comparison, the double sinusoidal structure exhibited superior spreading performance, which is attributed to the combined effects of its lower surface energy barrier and multi-directional capillary channels.

- (2) The laser-induced active defects promoted the rapid formation of a continuous and thickened reaction layer. This layer exhibited distinct functional gradient characteristics: the outer Ti₅Si₃/TiO₂ composite layer formed a strong metallurgical bond with the ceramic substrate, while the inner Cu₃Ti₃O layer demonstrated excellent compatibility with the AgCuTi filler metal, thereby effectively enhancing its wettability capability.
- (3) Compared to the single sinusoidal structure, the double sinusoidal structure exhibited higher surface roughness and surface area ratio values, resulting in a significantly larger active surface area. This accelerated the formation of the wettable Cu₃Ti₃O reaction layer, further enhancing wettability. Therefore, the double sinusoidal structure achieved remarkable wettability enhancement through the morphological advantages of low energy barriers and multi-directional capillary channels combined with the enhanced formation of a wettable reaction layer enabled by the large active surface area.
- (4) This double sinusoidal structure achieved a minimum wetting angle of ~70° under the conditions of 25μm amplitude and 100μm wavelength, marking a 36.4% decrease from the original surface. Moreover, the SiO₂f/SiO₂ joints under the same parameter attained 23MPa, reaching 4.6 times that of the original joint (5MPa). Therefore, the sinusoidal interfacial structure not only improves wettability but also enhances joint strength.

CRediT authorship contribution statement

Shuzhen Zhao: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Lixia Zhang:** Writing – review & editing, Supervision, Project administration. **Qing Chang:** Writing – review & editing. **Bo Zhang:** Validation, Investigation. **Zhan Sun:** Writing – review & editing. **Yongbo Xu:** Validation,

Investigation, Data curation. **Boyu Zhang:** Validation, Investigation. **Fengying Wang:** Investigation.

Declaration of Generative AI and AI-assisted technologies in the writing process

Generative AI and AI-assisted technologies should only be used in the writing process to improve the readability and language of the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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4



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